Table 2: Overview of data sets.

Data set	n	d	#CPs
CIFAR10 (Krizhevsky et al., 2009)	60,000	1,024	9
FashionMNIST (Xiao et al., 2017)	70,000	784	9
Gas (Vergara et al., 2012)	13,910	128	5
HAR (Anguita et al., 2013)	10,299	561	5
MNIST (Deng, 2012)	70,000	784	9

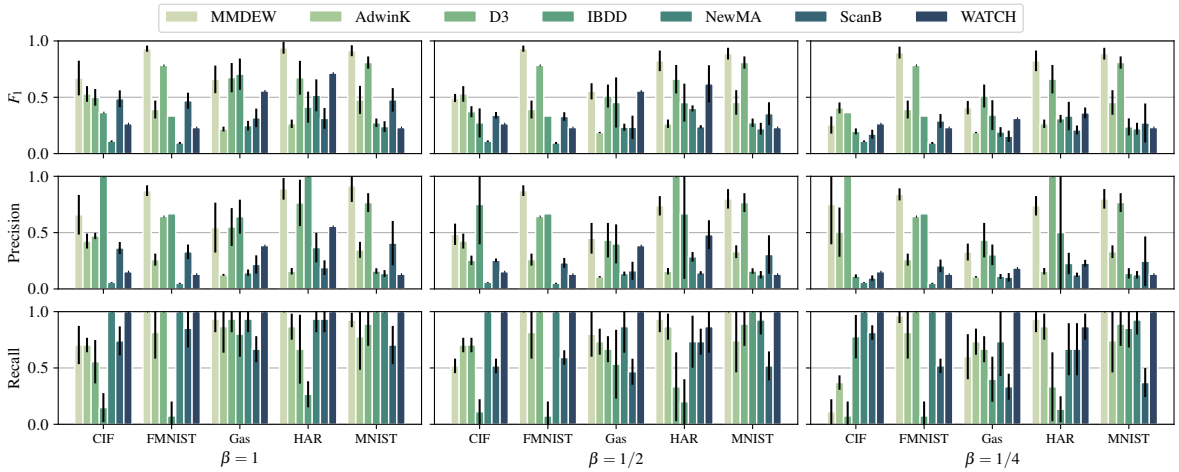

Figure 5: Average F_1-score, precision and recall. The bars show the standard deviation over 10 permutations of the data.

We run a grid parameter optimization per data set and algorithm and report the best result w.r.t. the F_1-score. We note that such an optimization is difficult to perform in practice—here one typically prefers approaches with fewer or easy-to-set parameters—but allows a fair comparison. Table 3 in Appendix C lists all the parameters we tested. We note that the grid parameter optimization allowed us to obtain better F_1-scores than the heuristics proposed in Keriven et al. (2020) for NEWMA and Scan B-statistics.

We exclude the squared time estimator of MMD due to its prohibitive runtime. For kernel-based algorithms (MMDEW, NEWMA, and Scan B-statistics) we use the Gaussian kernel $k(x, y) = \exp\left(-\gamma\|x - y\|^2\right)$ $(\gamma > 0)$ and set γ using the median heuristic (Garreau et al., 2018) on the first 100 observations. The Gaussian kernel is universal (Steinwart & Christmann, 2008; Szabó & Sriperumbudur, 2017) and allows, given enough data, to detect any change in distribution as a universal kernel on a compact domain is characteristic (Gretton et al., 2012, Theorem 5). We also supply the first 100 observations to competitors requiring data to estimate further parameters (IBDD, WATCH) upfront.

F_1-score, precision, and recall. We compute the precision, the recall, and the F_1-score, which are common to evaluate change detection algorithms (Li et al., 2019; Keriven et al., 2020; van den Burg & Williams, 2020; Faber et al., 2021). Specifically, for a fixed $\Delta_T \in \mathbb{N}_{>0}$, we proceed as follows. If a change is detected, and there is an actual change point within the Δ_T previous time steps, we consider it a true positive (tp). If a change is detected, and there is no change point within the Δ_T previous steps, we consider it a false positive (fp). If no change is detected within Δ_T steps of a change point, we consider it a false negative (fn). We count at most one true positive for each actual change point. With these definitions, the precision is $\mathrm{Prec} = \mathrm{tp}/(\mathrm{tp} + \mathrm{fp})$, the recall is $\mathrm{Rec} = \mathrm{tp}/(\mathrm{tp} + \mathrm{fn})$, and the F_1-score is their harmonic mean $F_1 = 2 \cdot (\mathrm{Prec} \cdot \mathrm{Rec}) \,/\, (\mathrm{Prec} + \mathrm{Rec})$. Note that, while some algorithms allow to infer where in the data a change happens, including the proposed MMDEW, we only evaluate the time at which they report a change, as all tested approaches allow reporting this value.

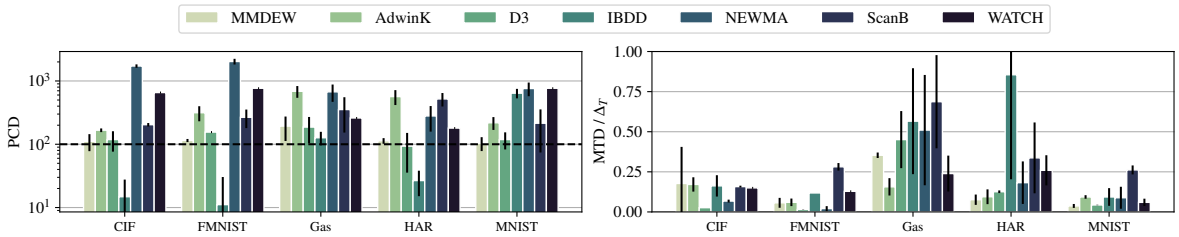

Figure 6: Average of percentage of changes detected (PCD) and of mean time to detection (MTD). The dashed line indicates the optimum for PCD. For MTD lower values are better.

Figure 5 shows our results. As Δ_T is an evaluation-specific parameter, we vary it relative to the average distance between change points by a factor $\beta > 0$: Given a data set of length N with n changes, we set $\Delta_T = \beta \cdot N/(n+1)$. For $\beta = 1$ (Δ_T is equal to the average number of steps between change points per respective data set), MMDEW achieves a higher F_1-score than all competitors on all data sets except for Gas, where it still obtains a competitive result. Throughout, the proposed algorithm obtains a good balance between precision and recall. Other approaches either have very low precision (for example, less than 20%), or an inferior recall and precision, down to a few exceptions. With a reduced β, that is, we allow only a shorter detection delay, the performance of all algorithms decreases on average. For $\beta = 1/2$, MMDEW achieves the best F_1 score also on four data sets, and, for $\beta = 1/4$ (the most challenging setting) on three of the tested data sets.

We conclude that the proposed method achieves very good results across all these experiments—especially when taking into account the fewer hyperparameters compared to the other approaches that we tested.

Percentage of changes detected and detection delay. To obtain a complete picture of the performance of MMDEW, we also report the "percentage of changes detected" (PCD), that is, the ratio of the number of reported changes and the number of actual change points, and its MTD on the data streams derived from real-world data. In our context, MTD coincides with the expected detection delay.

Figure 6 collects our results. For PCD, results closer to 100% are better. Here, MMDEW is on par with the closest competitors and consistently, that is, across all data sets, detects an approximately correct number of change points. D3, NEWMA, Scan B-statistics, and WATCH detect too many change points in all cases. This behavior is also reflected in their comparably large recall in Figure 5.

For MTD, lower values are better. Here, the classification-based D3 performs best in most of the cases. MMDEW performs a bit worse than D3 but better than the other algorithms on most data sets, with the Gas data set the major exception. As the experiments in Figure 5 show, a lower Δ_T tends to lead to a lower F_1-score of MMDEW. In other words, MMDEW tends to detect changes with some delay, but it detects them consistently.

6 Conclusions

We introduced a novel change detection algorithm, MMDEW, that builds upon two-sample testing with MMD, which is known to yield powerful tests on many domains. To facilitate the efficient computation of MMD, we presented a new data structure, which allows to estimate MMD with polylogarithmic runtime and logarithmic memory complexity. Our experiments on standard benchmark data show that MMDEW obtains the best F_1-score on most data sets. At the same time, MMDEW only has two parameters—the level of the statistical test and the choice of kernel. This simplifies the proposed algorithm's application in real-world use cases.

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